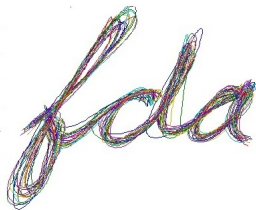


Functional data analysis

2. Random processes

Alfredas Račkauskas, Jurgita Markevičiūtė

Lecture notes, 2026



Outline of the lecture

① *Modelling functional sample*

② *Random processes*

Objectives:

- ▶ *to provide a basic probabilistic structure for how functional data are generated.*

Modelling functional sample I

A probability space is a triple $(\Omega, \mathcal{F}, P)$, where:

- ▶ Probability space $(\Omega, \mathcal{F}, \mathbb{P})$;
- ▶ Sample space Ω ;
- ▶ $\mathcal{F}$ is σ -algebra;
- ▶ $\mathbb{P}$ - probability measure.

Modelling functional sample II

Elements of a Probability Space

The three building blocks of a probability space can be described as follows:

- ▶ The sample space Ω , is a primitive concept, loosely defined as the set of all possible outcomes of the probabilistic experiment.
- ▶ The sigma-algebra $\mathcal{F}$ is the collection of all subsets of Ω to which we are able/willing to assign probabilities; these subsets are called events.
- ▶ The probability measure $\mathbb{P}$ is a function that associates a probability to each of the events belonging to the sigma-algebra $\mathcal{F}$.

Modelling functional sample III

Example:

$$(\Omega, \mathcal{F}, \mathbb{P})$$

Example (fair coin toss):

$$\Omega = \{H, T\}$$

$$\mathcal{F} = 2^\Omega = \{\emptyset, \{H\}, \{T\}, \Omega\}$$

$$\mathbb{P}(\{H\}) = \mathbb{P}(\{T\}) = \frac{1}{2}$$

So the probability space is:

$$(\{H, T\}, \{\emptyset, \{H\}, \{T\}, \{H, T\}\}, \mathbb{P})$$

Modelling functional sample IV

Sigma-Algebra

Let $\mathcal{F}$ be a set whose elements are subsets of Ω . Then, $\mathcal{F}$ is a sigma-algebra if and only if it satisfies the following axioms:

- ▶ $\mathcal{F}$ contains the sample space: $\Omega \in \mathcal{F}$.
- ▶ If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$ (where A^c is the complement of A , also denoted by $\Omega \setminus A$).
- ▶ If $\{A_n\}_{n \in \mathbb{N}}$ is a countable collection of elements of $\mathcal{F}$, then $\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{F}$.

It is possible to prove that these axioms imply:

- ▶ $\emptyset \in \mathcal{F}$.
- ▶ If $\{A_n\}_{n \in \mathbb{N}}$ is a countable collection of elements of $\mathcal{F}$, then $\bigcap_{n \in \mathbb{N}} A_n \in \mathcal{F}$.

Modelling functional sample V

Probability Measure

Let $\mathbb{P}$ be a function that associates a real number to each element of the sigma-algebra $\mathcal{F}$. Then, $\mathbb{P}$ is a probability measure if and only if it satisfies the following axioms:

- ▶ If $A \in \mathcal{F}$, then $\mathbb{P}(A) \geq 0$.
- ▶ $P(\Omega) = 1$.
- ▶ If $\{A_n\}_{n \in \mathbb{N}}$ is a countable collection of disjoint elements of $\mathcal{F}$ (i.e., $A_j \cap A_k = \emptyset$ if $j \neq k$), then:

$$P\left(\bigcup_{n \in \mathbb{N}} A_n\right) = \sum_{n \in \mathbb{N}} P(A_n).$$

Random processes

Here we recall the most important knowledge about random processes.

- ▶ Random process is a probabilistic model of a random experiment that evolves in time.
- ▶ For a fixed time t one assigns a random variable X_t thus obtaining a collection $(X_t, t \in \mathcal{T})$ of random variables indexed by a time parameter t .



- ▶ Formal definition of a random process is as follows.
- ▶ Let $(\Omega, \mathcal{F}, P)$ be a probability space.
- ▶ Let $\mathcal{T} \subset \mathbb{R}$ (we shall consider $\mathcal{T} = [0, 1]$, $\mathcal{T} = [0, T]$ or $\mathcal{T} = [0, \infty)$).

Definition

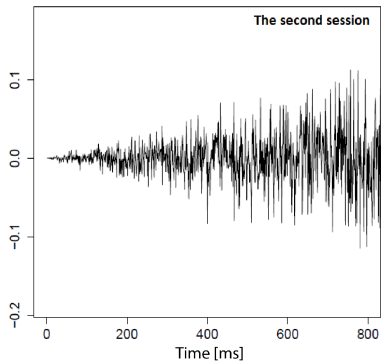
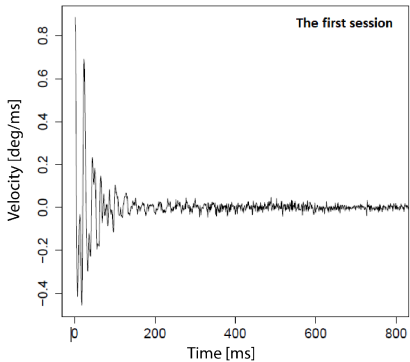
A *random process* (equiv. *stochastic process*) is a set of random variables $X := \{X(t), t \in \mathcal{T}\}$ defined on a probability space $(\Omega, \mathcal{F}, P)$.

- ▶ Random process $X = (X(t), t \in \mathcal{T})$ is a function of two arguments: $\omega \in \Omega$ and $t \in \mathcal{T}$:

$$X = (X(t, \omega) : t \in \mathcal{T}, \omega \in \Omega).$$

- ▶ If t is fixed, then $X(t) = (X(t, \omega), \omega \in \Omega)$ is a random variable.
 - ▶ If $\mathcal{L}_0(\Omega)$ is a set of all random variables defined on $(\Omega, \mathcal{F}, P)$, then $X(t) \in \mathcal{L}_0(\Omega)$ for every $t \in \mathcal{T}$.
 - ▶ Hence, X maps $\mathcal{T}$ into the set $\mathcal{L}_0(\Omega)$.
- ▶ If $\omega \in \Omega$ is fixed, then $X(\omega) = (X(t, \omega), t \in \mathcal{T})$ is a real valued function defined on $\mathcal{T}$.

- ▶ A plot of a stochastic process $X(t, \omega)$ as a function of $t \in \mathcal{T}$, for a fixed outcome $\omega \in \Omega$, is called a *sample path* of the process.
- ▶ When a stochastic process is plotted, it is a (hopefully "typical") sample path that is depicted.
 - ▶ If $\mathbb{R}^{\mathcal{T}}$ is a set of all functions $x : \mathcal{T} \rightarrow \mathbb{R}$, then $X(\omega) \in \mathbb{R}^{\mathcal{T}}$ for every $\omega \in \Omega$.
 - ▶ Hence, X maps Ω into the set $\mathbb{R}^{\mathcal{T}}$.



- In FDA we work with sample path of a random process $X = (X(t), t \in [0, 1])$, which customary one denotes by $x = (x(t), t \in [0, 1])$. Hence,

$$x = X(\omega), \quad x(t) = X(t, \omega) \quad \text{for an } \omega \in \Omega.$$

- The functional sample of size n , say, $x_1, \dots, x_n$, we interpret then as sample paths of n random processes, say, $X_1, \dots, X_n$. Hence,

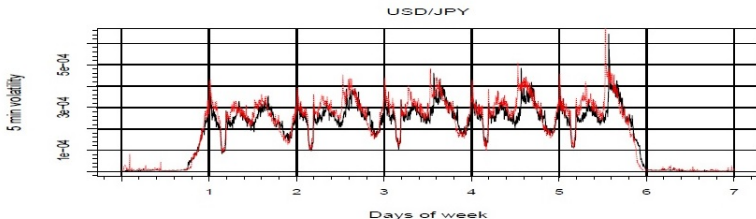
$$x_i = X_i(\omega), \quad x_i(t) = X_i(t, \omega) \quad \text{for an } \omega \in \Omega.$$

$$i = 1, \dots, n.$$

Remark

The processes $X_1, \dots, X_n$ can be obtained by segmentation of one random process $X = (X(t), t \in [0, T])$:

- ▶ choosing δ so that $n = T/\delta$, and taking
 $X_i(t) := X((i-1)\delta + \delta t), \quad t \in [0, 1], \quad i = 1, \dots, n.$



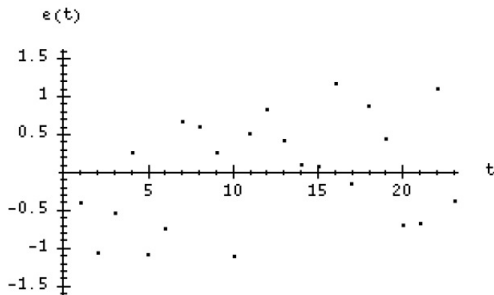
Example 1

A basic, very important stochastic process in discrete time is discrete white noise $(e(t), t \in \mathbb{Z})$, where

$$\dots, e(-1), e(0), e(1), \dots$$

are random variables such that

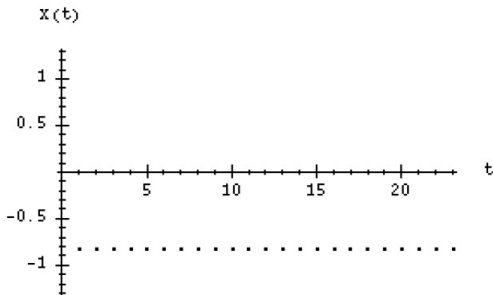
$$E(e(t)) = 0, \text{ var}\{e(t)\} = \sigma^2 \text{ and } \text{cov}\{e(s), e(t)\} = 0 \text{ for } s \neq t.$$



Discrete Gaussian white noise with variance $\sigma^2 = 1$.

Example 2

Consider the process $X(t) = \xi$ for $t \in \mathbb{Z}$, where ξ is a single random variable. A sample path of this process is just the horizontal sequence of equidistant points $(t, \xi(\omega)) \in \mathbb{R}^2, t \in \mathbb{Z}$. In other words, the process values are completely dependent on each other: If you know one process value, you know all of them!



The process with $\xi \sim \mathcal{N}(0, 1)$.

Finite dimensional distributions

- ▶ Consider a stochastic process $X = (X(t), t \in \mathcal{T})$.
- ▶ Usually, in both theory and applications, process values $X(s)$ and $X(t)$, at different times s and t , depend more of one another than they do in Example 1, but less than in Example 2.
- ▶ The analysis of stochastic processes is much about the study of that dependence.
- ▶ The analysis of the dependence structure of a stochastic process requires, in principle, a complete knowledge of the finite dimensional distributions of the process.

Finite dimensional distributions

- First-order distribution functions of X are :

$$F_t(u) = P(X(t) \leq u) = P(\{\omega : X(t, \omega) \leq u\}), \quad u \in \mathbb{R}, \quad t \in \mathcal{T}.$$

- Second-order distribution function of X are:

$$F_{t_1, t_2}(u_1, u_2) = P(X(t_1) \leq u_1, X(t_2) \leq u_2), \quad u_1, u_2 \in \mathbb{R}; \quad t_1, t_2 \in \mathcal{T}.$$

- n 'th-order distribution function of X are:

$$F_{t_1, \dots, t_n}(u_1, \dots, u_n) = P(X(t_1) \leq u_1, \dots, X(t_n) \leq u_n), \\ u_1, \dots, u_n \in \mathbb{R}, \quad t_1, \dots, t_n \in \mathcal{T}.$$

Definition

The collection

$$\mathcal{F}^{(X)} := \{F_{t_1, \dots, t_n} : t_1, \dots, t_n \in \mathcal{T}; n \geq 1\}$$

is called a family of finite dimensional distributions of a stochastic process $X = (X(t), t \in \mathcal{T})$.

Definition

Two random processes X and Y are identically distributed (denoted $X \stackrel{\mathcal{D}}{\sim} Y$) if their corresponding finite dimensional distributions are equal:

$$X \stackrel{\mathcal{D}}{\sim} Y \text{ if } \mathcal{F}^{(X)} = \mathcal{F}^{(Y)}.$$

- However, in practice these distributions can seldom be calculated. Instead of an exact analysis, one must therefore often feel satisfied with approximations. Computer simulation of processes are often important aids, to find such approximations.

First- and second-order moments of random process

Definition

Assume that $E|X(t)| < \infty$ for each $t \in \mathcal{T}$.

- ▶ The function

$$\mu_X(t) = E(X(t)), \quad t \in \mathcal{T},$$

is called the mean function (or simply the mean) of X .

- ▶ The random process

$$\tilde{X}(t) = X(t) - \mu_X(t), \quad t \in \mathcal{T},$$

is called the centred (or compensated) random process, corresponding to X .

- Recall that covariance of two random variables ξ and η , both having the second moment, is defined as

$$\text{cov}(\xi, \eta) = E(\xi - E(\xi))(\eta - E(\eta)) = E(\xi\eta) - E(\xi)E(\eta).$$

Definition

Assume that $E(X^2(t)) < \infty$ for each $t \in \mathcal{T}$. In this case we say that the process X is *of second order*.

- The function

$$\mathcal{K}_X(t, s) = \text{cov}(X(t), X(s)), \quad t, s \in \mathcal{T},$$

is called the *covariance function* (or simply the covariance) of the random process X .

Definition

If X, Y are two random processes with $E(X^2(t)) < \infty, E(Y^2(t)) < \infty$ for each $t \in \mathcal{T}$, then

$$\mathcal{K}_{X,Y}(t,s) = \text{cov}(X(t), Y(s)), \quad t, s \in \mathcal{T},$$

is called the *cross covariance function* of the random processes X, Y .

We will write either

$$X \sim \text{Sp}(\mu, \mathcal{K})$$

or

$$(X(t), t \in \mathcal{T}) \sim \text{Sp}(\mu, \mathcal{K})$$

to denote a second order stochastic process with mean μ and covariance $\mathcal{K}$.

If μ_X is a constant and $R_X(t, s) = \rho_X(t - s)$, $t, s \in \mathcal{T}$, we say that the random process $(X(t), t \in \mathcal{T})$ is wide-sense stationary.

Definition

A process $(X(t), t \geq 0)$ has *independent increments* if, for each choice of $n \in \mathbb{N}$ and $0 \leq t_0 < t_1 < \cdots < t_n$, the random variables

$$X(t_n) - X(t_{n-1}), \dots, X(t_2) - X(t_1), X(t_1) - X(t_0)$$

are independent.

Definition

A process $(X(t), t \geq 0)$ has *stationary increments* if, for every $h \geq 0$,

$$X(t+h) - X(t) \stackrel{\mathcal{D}}{\sim} X(h) - X(0).$$

Definition

A process $(X(t), t \geq 0)$ is a *Lévy process* if increments are independent and stationary.

Gaussian processes

Definition

A stochastic process $X = (X(t), t \in \mathcal{T})$ is Gaussian with mean function $\mu = (\mu(t), t \in \mathcal{T})$ and covariance function $\mathcal{K} = (\mathcal{K}(s, t), s, t \in \mathcal{T})$, denoted as

$$X \sim \text{GSp}(\mu, \mathcal{K}),$$

if and only if for any p time points, $t_j, j = 1, 2, \dots, p$, the random vector $[X(t_1), \dots, X(t_p)]'$ follows a multivariate normal distribution $\mathcal{N}(\boldsymbol{\eta}, \boldsymbol{\Gamma})$, where

$$\boldsymbol{\eta} = [\mu(t_1), \dots, \mu(t_p)]' \quad \text{and} \quad \boldsymbol{\Gamma} = [\mathcal{K}(t_i, t_j), i, j = 1, \dots, p].$$

Hence, if the process $(X(t), t \in \mathcal{T})$ is Gaussian, then the finite dimensional distributions are determined by the expectation function $\mu_X(t)$ together with the covariance function $\mathcal{K}_X(s, t)$.

Wiener process

A stochastic process $(W(t), t \geq 0)$ is called a Wiener process if

- i) $W(0) = 0$;
- ii) $(W(t), t \geq 0)$ is a process with independent and stationary increments;
- iii) $W(t) \sim \mathcal{N}(0, \sigma^2 t)$, for $t > 0$;
- iv) all trajectories of the process are continuous.

If $\sigma^2 = 1$ the W is a standard Wiener process.

For $t, s \geq 0$,

$$\mu_W(t) = E(W(t)) = 0, \quad \mathcal{K}_W(t, s) = \sigma^2 \min\{t, s\}$$

Restriction of the Wiener process $W = (W(t), t \geq 0)$ on a set $[0, 1]$ is again called a Wiener process and denoted $W = (W(t), t \in [0, 1])$.

Theorem

A Wiener process $(W(t), t \geq 0)$ is a Lévy process with $W(0) = 0$ and $W(t+h) - W(t) \sim \mathcal{N}(0, \sigma^2 h)$ for $t \geq 0$ and $h \geq 0$.

Self-similar processes are stochastic processes that are invariant in distribution under suitable scaling of time and space.

Theorem

A Wiener process $(W(t), t \geq 0)$ is self-similar in the sense that for each $a > 0$

$$(W(at), t \geq 0) \stackrel{\mathcal{D}}{\sim} (a^{1/2} W(t), t \geq 0).$$

Simulation of Wiener process

Algorithm:

1. Divide the time interval $[0, T]$ into n subintervals of length $\Delta t = T/n$.
2. Set $W(0) = 0$.
3. Generate next states according to

$$W(t + \Delta t) = W(t) + Z\sqrt{\Delta t},$$

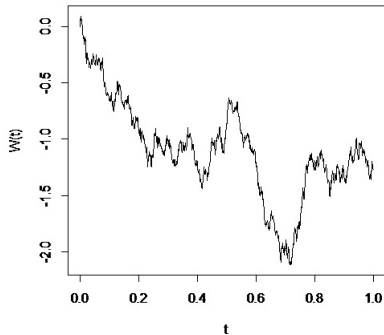
where $Z \sim \mathcal{N}(0, 1)$, or

4. find cumulative sum of

$$W(n\Delta t) = \sum_{k=1}^n Z_k \sqrt{\Delta t}$$

and $Z_1, Z_2, \dots \stackrel{iid}{\sim} \mathcal{N}(0, 1)$.

```
n<-1000; T. <-1  
delta<-T/n  
W<-cumsum(c(0,rnorm(n,0,1)*sqrt(delta)))  
plot(seq(0,T,delta),W,type="l",  
xlab="t",ylab="W(t)",xlim=c(0,T+delta))
```



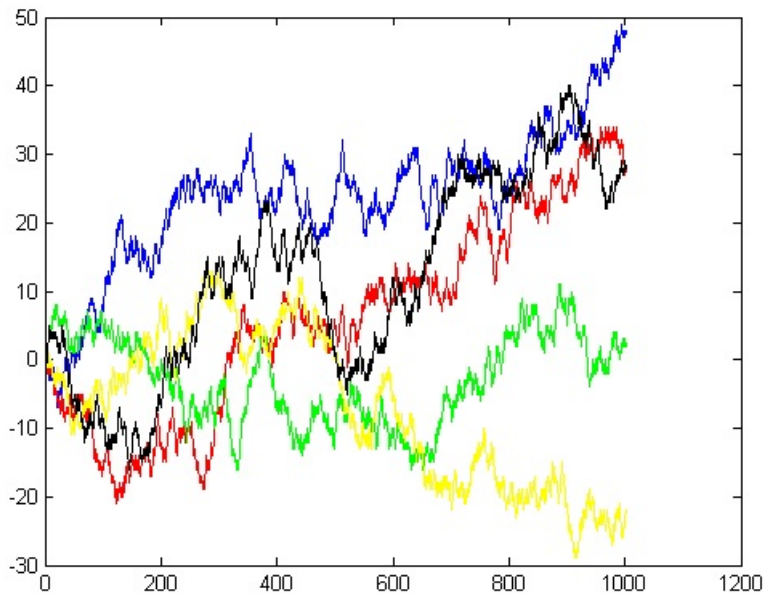
This simulation algorithm is based on invariance principle.

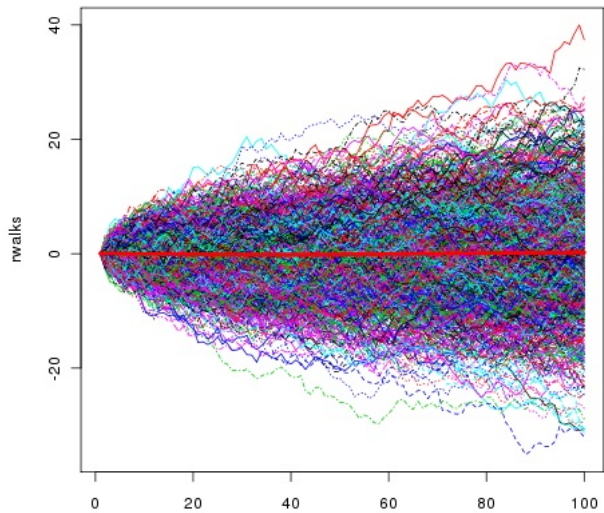
Invariance principle

If $X_1, X_2, \dots$ are iid random variables such that $E(X_i) = 0$, $\text{var}(X_1) = 1$, then the process

$$Z_n(t) := n^{-1/2} \sum_{k=1}^{[nt]} X_k + n^{-1/2}(nt - [nt])X_{[nt]+1}, \quad t \in [0, 1],$$

converges (in distribution) to the standard Wiener process $W = (W(t), t \in [0, 1])$.





- ▶ Another effective method to simulate Wiener process by using Karhunen-Loève expansion:

$$W_t = \sqrt{2} \sum_{k \geq 0} \gamma_k \frac{2}{(2k+1)\pi} \sin \left(\left(k + \frac{1}{2} \right) \pi t \right),$$

where $\{\gamma_k\}$ are i.i.d. Gaussian random variables with mean zero and variance one.

Processes related to Wiener

Brownian bridge

If $W = (W(t), t \in [0, 1])$ is a standard Wiener process, then the process $B = (B(t), t \in [0, 1])$, where

$$B(t) = W(t) - tW(1)$$

is called a standard Brownian bridge.

Ornstein-Uhlenbeck process

If $W = (W(t), t \in [0, 1])$ is a standard Wiener process, then the process $U = (U(t), t \in [0, 1])$, where

$$U(t) = \mu(t) + \frac{\sigma}{2\sqrt{\theta}} e^{-\theta t} W(e^{2\theta t})$$

is called an Ornstein-Uhlenbeck process with parameters (μ, σ, θ) .

- ▶ $E(U(t)) = \mu(t)$;
- ▶ $\text{cov}(U(t), U(s)) = \frac{\sigma^2}{2\theta} e^{-\theta|t-s|}$.

- If $W_1 = (W_1(t), t \geq 0)$ and $W_2 = (W_2(t), t \geq 0)$ we defined the Wiener process $(W(t), t \in \mathbb{R})$ by setting

$$W(t) = W_1(t)\mathbf{1}(t \geq 0) + W_2(-t)\mathbf{1}(t < 0).$$

- Let $(W(t), t \in \mathbb{R})$ be a Wiener process and $f : \mathbb{R} \rightarrow \mathbb{R}$ a function such that

$$\int_{-\infty}^{\infty} f^2(s)ds < \infty.$$

- It is not too difficult to show that the following stochastic integral is well-defined as

$$\int_{-\infty}^{\infty} f(s)dW(s) = P\text{-}\lim_{n \rightarrow \infty} \sum_{k=-s(n)}^{s(n)} f(k/n) [W((k+1)/n) - W(k/n)].$$

Here the summation limit $s(n)$ should satisfy
 $\lim_{n \rightarrow \infty} s(n)/n = \infty.$

Theorem

The stochastic process

$$X(t) = \int_{-\infty}^{\infty} f(t+s)dW(s) \quad t \in \mathbb{R}$$

is Gaussian and stationary, with expectation and covariance function

$$\mu_X(t) = 0 \quad \text{and} \quad \mathcal{K}_X(\tau) = \int_{-\infty}^{\infty} f(s)f(\tau+s)ds.$$